

Handbook

COOPERATIVE EDUCATION

(Co-op) Program at MSFEA

MSFEA Career Development Center



**AMERICAN
UNIVERSITY** OF BEIRUT

**MAROUN SEMAAN FACULTY OF
ENGINEERING & ARCHITECTURE**

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| Co-op Definition

Cooperative education, also known as co-op, is a college-held program in which students combine their studies with one or more official paid work terms within their area of study. The difference between a co-op work term and an internship is that internships are typically short in duration (one or two months) and can be unpaid, while co-op terms typically have a longer duration (at least four months) and are paid. According to the U.S. Department of Education, students with co-op experience earn almost 8-9% more within ten years of graduation in comparison to students with a graduate degree.

| Motivation and Need

A co-op experience enables students to immerse themselves in an industry and better understand their study area. They come back to the university with a better perception of what they learn and can easily spot the applications of their learning experience. Moreover, a paid co-op gives the students a better sense of responsibility which makes their experience at the company more meaningful. The employers would also benefit from access to talent and early recruitment in addition to becoming a part of MSFEA's Corporate Partnership program.

Co-op programs have been shown to significantly increase student interest in learning and academic performance when they return to campus, and have been shown to lead to more satisfying career outcomes ¹(Blair et al, 2004; ²Boudreau and Marx, 2019)

| Co-op Program at MSFEA

Components

The co-op program at MSFEA consists of two components: Career Development Preparation and the Cooperative Education Experience itself.

Career Development Preparation: The MSFEA Career Development Center (CDC) is developing the following 0-credit Career Development Modules, which will be available asynchronously via Moodle:

- Resume and Cover Letter writing
- Interview preparation
- Work placement search
- Workplace environment and skills

These modules are required of co-op students before they are allowed to apply for co-op positions. In case the online modules are not ready for the first cohort, they will be replaced by workshops.

The CDC will also support students in their co-op job search and applications by offering students one-to-one consultations, if needed (e.g., to review their resumes, provide career counseling, and hold mock interviews).

¹ Blair, B.F., Millea, M. and Hammer, J. (2004), The Impact of Cooperative Education on Academic Performance and Compensation of Engineering Majors. *Journal of Engineering Education*, 93: 333-338. <https://doi.org/10.1002/j.2168-9830.2004.tb00822.x>

² Boudreau, Kevin and Marx, Matt, From Theory to Practice: Field Experimental Evidence on Early Exposure of Engineering Majors to Professional Work (June 20, 2019). Available at SSRN: <https://ssrn.com/abstract=3271144> or <http://dx.doi.org/10.2139/ssrn.3271144>

Cooperative Education Experience: Students who join the co-op program should complete one job placement experience with a minimum duration of 6 months. The 3-credit course FEAA 500 is associated with this Cooperative Education Experience. Students may participate in (and employers may propose) multiple 6-months co-op experiences (consecutive or not).

The objectives of the co-op experience are for students to:

- Apply and enhance their technical skills
- Learn and practice professional and communication skills
- Be introduced to the industry and work environment
- Gain deeper understanding of the field they are working in and its qualification requirements
- Interact and consult with supervisors and colleagues in a professional environment
- Make better career decisions and field selections after returning to the university
- Build connections with professionals in their field and increase their chances of employment after graduation.

People

The co-op program at MSFEA involves co-op students, planning to or already enrolled in the co-op program, co-op employers, planning to or already participating in the co-op program, and is administered by a co-op committee that consists of a Chair, a CDC Advisor and Academic Advisors from participating departments.

Policies

Optional or Required? Optional, open to undergraduate students (it could also be open to graduate students if allowed by the department).

Hours of Work and Salary: Students are required to work in paid full-time jobs during their co-op (a minimum of 35 to 40 hours a week). Students should be paid for their work (higher than a stipend but slightly below a fresh graduate salary). MSFEA can provide salary guidelines, but salaries are set by the employer.

Courses while on Co-op: Enrolling in courses during a co-op is not allowed. In some cases, engineering students may be allowed to take a Final Year Project (FYP) course or Research based course(499), if allowed by their department and approved by their co-op academic advisor and the other FYP group members in the case of FYP. If allowed, the FYP or 499 coursework should not intervene with the working hours.

Number of Credits: The co-op experience is associated with a 3-credit Pass/Fail FEAA 500 course. Whether the course counts or not towards academic credit, and whether it is considered as equivalent to a free elective or a technical elective course is decided by individual departments. Students maintain their full-time student status during the co-op experience.

Tuition or Co-op Fees: Regular tuition fees are charged for the 3 credits of the FEAA 500 course. Students who take and Pass FEAA 500 will no longer be required to take the 0-credit Approved Experience/Professional Training course in the department. The latter is currently billed as 1 credit.

Number of Co-ops, Duration of each Co-op, and Co-op Certification: Students who join the co-op program should complete one job placement experience with a minimum duration of 6 months. They are allowed to complete additional co-ops (consecutive or not) if they wish, and if approved by their department and co-op advisor(s). If a student enrolls in the co-op course (FEAA 500) more than once, the consideration of the course towards academic credit (number and type) for each enrollment is decided by individual departments. The 6-months co-op should

typically coincide with the Summer + Fall semesters or the Spring + Summer semesters. The co-op experience appears as a course on the academic transcript.

Timing of Co-op and Return to Campus after Co-op: The co-op experience should take place during the 3rd or 4th year of study for engineering and graphic design students and during the 4th or 5th year for architecture students. Students are required to return to campus for at least one term following their final co-op to take courses prior to graduating.

Preparation and Support for Co-op: Students are provided with support and preparation by co-op CDC and academic advisors before and during their co-ops. The provided support and preparation include, among others: co-op Info Sessions; co-op advising weeks; on-demand meetings with the co-op CDC Advisor to review resumes, provide career counseling, and hold mock interviews; co-op career development modules; access to co-op employers and job opportunities; follow-up and feedback during and after the co-op; and faculty mentorship and assessment.

Minimum GPA and Other Student Eligibility Requirements: For students to participate in the co-op program, they must be an undergraduate student in MSFEA, have a minimum GPA of 3.3 (i.e., B+ or 79-82), and have completed at least 90 credits (120 credits for architecture students). Stricter minima may be set by individual departments or employers. Additional student eligibility requirements may be set by individual departments such as specific departmental course pre-requisites. The co-op program could also be open to graduate students if allowed by their department.

The student applications are reviewed by their department's co-op academic advisor, who makes accept or reject decisions in consultation with their department or the Undergraduate Committee of their department (if present). Once accepted to the co-op program, students should complete the co-op career development modules to be allowed to apply to co-op positions and enroll in the co-op course.

Requirements to Pass a Co-op Course: To pass the co-op course, students must satisfy the following requirements: submit a one-page progress report; have a successful midterm site visit (in-person or online) by the co-op advisor(s); obtain a favorable Final Student Performance Evaluation Form from their job supervisor; submit a final report (a three page memo describing the work completed plus appendices providing work samples); and complete a Student Self Evaluation, Reflection and Feedback Form. Additional requirements may be set by individual departments, such as submission of a portfolio of work produced for architecture and graphic design students.

Program Length and Graduation Date: Students may be permitted to take multiple overloads to complete the co-op without delay in graduation. Such cases require careful planning of the course load and schedule in advance, as well as approval of their department or the Undergraduate Committee of the department (if present) on a case-by-case basis. However, in some cases, participation in the co-op program might result in a delay of the date of graduation by one term (for one co-op).

Refer to the section Information for Students for more details on the student application and admission processes, eligibility requirements, and the co-op course requirements, and to Appendix 1 for the FEAA 500 course syllabus.

I Information for Students

Student Application and Admission Process

Students can apply to the co-op program starting at the beginning of the Spring of their 2nd year. They should apply at least one year before the intended start date of their co-op position, which should take place during their 3rd or 4th year for engineering and graphic design students and during their 4th or 5th year for architecture students. To apply to the co-op program, students should submit the [Co-op Student Application Form](#) and meet with their co-op Academic Advisor during the co-op advising week to discuss their application, eligibility requirements, career interests and course schedule. For acceptance into the co-op program, they must satisfy the MSFEA student eligibility requirements (as listed below) and their department's additional eligibility requirements (if any – listed in the Departmental Policies). The applications are evaluated and decisions are taken by the co-op academic advisor of the department, in consultation with the department or department Undergraduate Committee (if present). Once accepted, students should complete the required co-op career development modules to be allowed to apply to co-op positions. After they complete the modules, they can apply to the co-op positions advertised by the Career Center, or seek approval of the co-op CDC and academic advisors for co-op job opportunities they find themselves. In the latter case, the student should submit a [Proposal of Cooperative Education and Experiential Learning Form](#) and contact the co-op CDC Advisor to ask for information on how to proceed. If a student receives an offer from an employer and accepts it, s/he should then sign the Co-op Joint Agreement Form along with the company/employer and CDC Advisor. A deadline is associated with each step of the process, for both “Summer + Fall” and “Spring + Summer” co-op positions, as summarized in Figure 1.

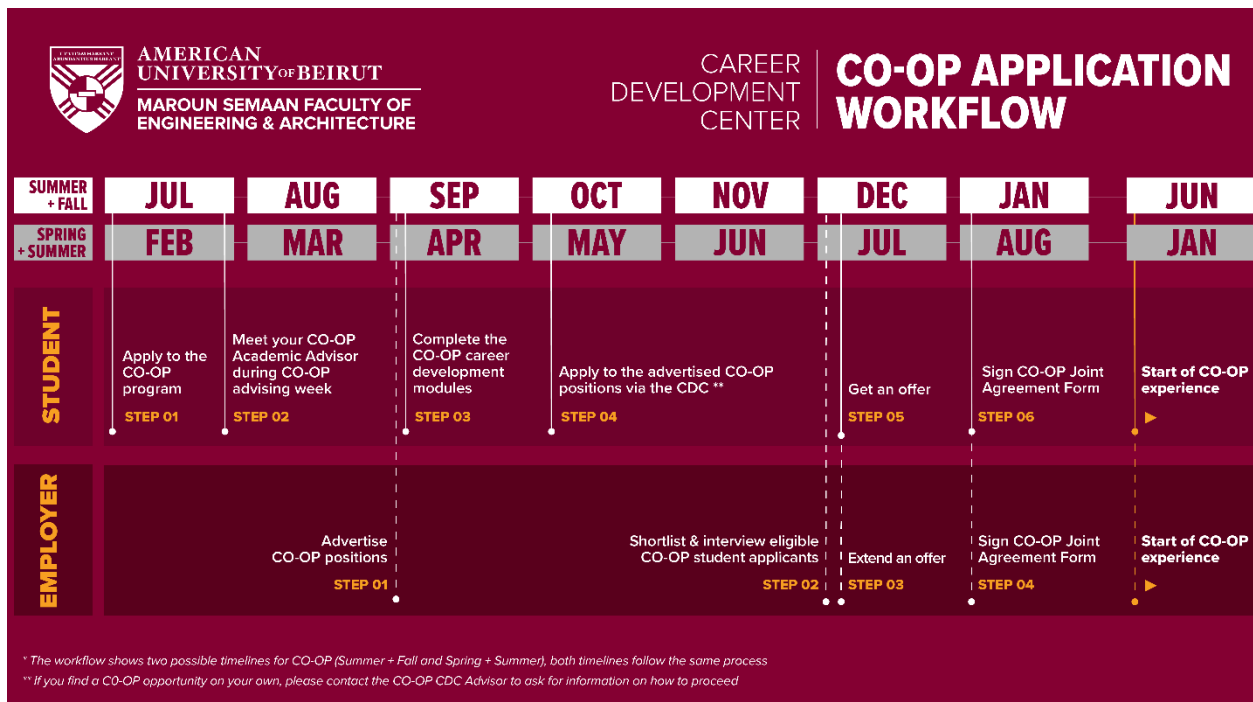


Figure 1 Workflow for students and employers applying to the co-op program.

Co-op course

If a student receives an offer and accepts it, s/he should enroll in the FEAA 500 course in both semesters of the co-op position (i.e., in Summer and Fall or in Spring and Summer). In the first semester of co-op, the student should register FEAA 500 (3-credits) and pay its tuition fees. In the second semester of co-op, the student should register FEAA 500A (3-credits), but with no additional tuition fees. Full-time student status is maintained in both semesters of the co-op experience. During and after the co-op, the student should submit a number of deliverables (forms and reports), which will be used as a basis for the evaluation by the co-op advisors. Students who have a satisfactory performance at the end of their first co-op semester receive a grade of "In Progress, PR" on FEAA 500. The final co-op grade of Pass or Fail is assigned on FEAA 500A at the end of the second co-op semester.

Student Eligibility Requirements

To be eligible to participate in the co-op program and enroll in the FEAA 500 course, students should satisfy the following criteria:

1. **Degree:** undergraduate student in MSFEA (it could also be open to graduate students if allowed by the department).
2. **Min GPA:** 3.3 (B+) required for all students at the time of application to the co-op program and when enrolling in the FEAA 500 course.
3. **Minimum number of credits:** the student should have completed a minimum of 90 credits (120 credits for Architecture students) when enrolling in the FEAA 500 course (not at the time of application to the co-op program).
4. **Career development workshops and modules offered by the CDC:** the student should have successfully completed the required modules/workshops before they can apply to co-op positions and enroll in the FEAA 500 course.
5. **Additional requirements set by individual departments:** individual departments can set a higher minimum GPA, a higher minimum number of credits, specific pre-requisite courses and additional requirements for their students. These requirements are decided and set internally by each department (refer to Departmental Policies).

Student Attitude, Behavior and Responsibilities at Work

Students should maintain a highly professional attitude and behavior at work. Punctuality and attendance at the required working hours are essential. Good relations with supervisors and colleagues should be a major consideration to project the proper attitude toward others. Students should remember that they are representing AUB during the co-op. During the co-op, students should also regularly meet with their employer/supervisor to seek their input and feedback.

Moreover, the co-op advisor(s) will liaise with the employer to conduct a midterm site visit (in person or online). Before the site visit, the student should have submitted the co-op progress report. During the site visit, the co-op advisors ask the employer about the student's performance, and the co-op advisors, employer and student discuss the student's experience and the employer's evaluation of the student's performance. Additional site visits may be scheduled, if needed, to enhance communication between university, employer and student.

Student Deliverables and Deadlines

Students are responsible for maintaining an individual co-op experience file that will allow for the evaluation of the co-op experience. The student deliverables and timeline for submissions in FEAA 500 are listed below.

Student deliverables during the co-op experience include:

1. Notice of Arrival Form (submitted through Moodle), to be sent promptly during the first week of the co-op.
2. One-page Co-op Progress Report, to be submitted at the end of the 3rd month.

Student deliverables at the end of the co-op experience include:

3. Co-op Final Report (a three-page memo describing the work completed plus appendices providing work samples).
4. [Student Self Evaluation, Reflection, and Feedback Form](#).
5. Additional requirements may be set by individual departments (refer to Departmental Policies).

Student Evaluation

Students who have made satisfactory progress in the first semester of their co-op experience will receive a grade of “In Progress, PR” on the FEAA 500 course. The final Pass or Fail grade on the Cooperative Education and Experiential Learning is assigned to the FEAA 500A course, at the end of the second semester. The co-op CDC and academic advisors will evaluate the various deliverables. The grade will be provided based on the following criteria:

1. Completion of the co-op experience file as described above.
2. Evaluation of the quality and relevance of the professional experience acquired.
3. Quality of the Co-op Progress and Final Reports.
4. Satisfactory student performance evaluation by the employer during site visit by the co-op advisor(s).
5. Satisfactory Final Student Performance Evaluation by the employer. The evaluation should reflect a performance and attitude deemed worthy by the employer.
6. Completion of Student Self Evaluation, Reflection, and Feedback on the co-op experience.
7. Additional criteria may be set by individual departments (refer to Departmental Policies).

Students may be requested to revise or repeat any of the deliverables in case they do not meet department expectations.

I Information for Employers

Employment Positions Requirements

Co-op positions offered to MSFEA students should satisfy the following criteria:

1. **Established company** known or renowned for offering meaningful training experiences.
2. **History of relationship** with AUB and MSFEA (preferred).
3. **Quality and relevance of the proposed experience:** evaluated by the co-op CDC and academic advisors based on a detailed job description to ensure that it involves a minimum level of tasks and responsibilities.
4. **Paid positions:** positions should be remunerated, at a level higher than a stipend and slightly below a full-time fresh graduate salary.
5. **Duration of position:** position should require 35-40 hours of work per week for a period of 6 months.
6. **Additional requirements set by individual departments:** individual departments can set additional requirements for companies and co-op positions for their students. These requirements are decided and set internally by each department (refer to Departmental Policies).

Employer Application Process

Companies and employers interested in advertising one or more co-op positions for MSFEA students should submit a [Co-op Employer Application Form](#), which includes a detailed job description of the co-op position(s). If the position(s) is (are) approved by the relevant department and co-op academic advisor, the employer will then receive a list of eligible co-op student applicants from the co-op CDC advisor. The employer can then shortlist and interview the students and decide whether to extend or not an offer. If an offer is extended and accepted by the student, the company/employer should then sign the Co-op Joint Agreement Form along with the student and CDC Advisor. A deadline is associated with each step of this process, for both “Summer + Fall” and “Spring + Summer” co-op positions, as summarized in Figure 1.

Employer Responsibilities During the Co-op Experience

During the co-op position, the employer/supervisor should regularly meet with the student, and be available for a midterm site visit by the co-op advisor(s) (in-person or online). During the site visit, the co-op advisor(s) ask the employer about the student’s performance, and the co-op advisor(s), employer and student meet to discuss the student’s experience and the employer’s evaluation of the student’s performance. At the end of the co-op position, the employer/supervisor should evaluate the student again and submit to the co-op CDC Advisor a [Final Student Performance Evaluation Form](#), which also states the dates of the co-op position and a brief description of the nature of the work performed by the student. The employer should finally submit a [Co-op Employer Experience Feedback Form](#).

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